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1. You are given the two DNA sequences. Sequence A: ACCGTGC and Sequence B: ACGTC.
 - (a) Perform a global alignment of the two sequences using the Needleman–Wunsch algorithm with the following scoring scheme: Match = +2, Mismatch = -1, Gap = -2. Write down the final alignment and the alignment score. [4]
 - (b) Perform a local alignment of the same sequences using the Smith–Waterman algorithm with the same scoring scheme. Show the highest-scoring local alignment and its score. [4]
 - (c) Briefly explain why the alignments differ between global and local methods in this case. [2]
2. (a) You are given the following distance matrix among four species (A, B, C, D):

	A	B	C	D
A	0	4	6	6
B	4	0	6	6
C	6	6	0	4
D	6	6	4	0

- i. Using the **UPGMA algorithm**, construct the phylogenetic tree step by step. Show which clusters are merged at each step and the new distances after merging. Indicate the height (ultrametric distance) at each merge. [6]
 - ii. Draw the final rooted phylogenetic tree (with branch lengths). [2]
- (b) What is an ultrametric tree? How is that relevant in evolutionary understanding? [2]
3. (a) Explain small world effect in Biological Networks with its implications. [3]
- (b) You are trying to predict cancer based on gene expression data. One difficulty that you are facing is that you have 1000 samples from cancer patients whereas only 100 samples from normal patients. This is creating an imbalance in your data. Explain how neural networks can be used to tackle this problem. [3]
- (c) You are trying to predict the function of a protein given its sequence. You have formulated the problem as supervised learning. As features what are the options do you have? [4]
4. (a) HMMs are often applied to gene finding, where exons and introns can be modeled as different states. Explain how transition probabilities and emission probabilities help distinguish between exon and intron regions. [6]
- (b) What is the space and runtime complexity of the *viterbi* algorithm? [2]
- (c) Can Baum-Welch algorithm guarantee optimal model and parameters? What is the solution to this problem? [2]